Complex Analysis Seminar

I. Some review of Basic theorems

1. Homomorphic functions and Cauchy integral theorems

Def I.1.1. $f: \mathbb{C} \to \mathbb{C}$ say $f(x,y) = u(x,y) + iv(x,y)$. $f|_\Omega$ is holomorphic iff $f'(z)$ exists on the open set Ω.

Def I.1.2. Two differential operations: $\frac{\partial}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} + \frac{1}{i}\frac{\partial}{\partial y}\right)$, $\frac{\partial}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} - \frac{1}{i}\frac{\partial}{\partial y}\right)$

Thm I.1.3. $f = u + iv$ is defined on a open set Ω. Then f is holomorphic on Ω if and only if (u, v) are differentiable and $\frac{\partial f}{\partial z} = f'(z)$. $\frac{\partial f}{\partial \bar{z}} = 0$

proof: (skipped)

Thm I.1.4. Given a power series $\sum_{n=0}^{\infty} a_n z^n$, let $\frac{1}{R} = \overline{\lim_{n\to\infty}} |a_n|^{\frac{1}{n}}$

(i) $|z| < R$. the series converges absolutely

(ii) $|z| > R$. the series diverges

proof: (skipped).

Thm I.1.5 $f(z) = \sum_{n=0}^{\infty} a_n z^n$ defines a holomorphic function in its disc of convergence.

Moreover, $f'(z) = \sum_{n=0}^{\infty} n a_n z^{n-1}$, f' shares the same disc of convergence with f

Pf: Let $|z_0| < r < R$ $\frac{1}{R} = \overline{\lim_{n\to\infty}} |a_n|^{\frac{1}{n}}$. $g(z) = \sum_{n=0}^{\infty} n a_n z^{n-1}$, $S_N(z) = \sum_{0\le n\le N} a_n z^n$. $E_N(z) = f(z) - S_N$

$$\frac{1}{h}\left(f(z_0+h) - f(z_0)\right) - g(z_0) = \left[\frac{1}{h}\left(S_N(z_0+h) - S_N(z_0)\right) - S_N'(z_0)\right] + \left(S_N'(z_0) - g(z_0)\right) + \frac{E_N(z_0+h) - E_N(z_0)}{h}$$

$$\left|\frac{E_N(z_0+h) - E_N(z_0)}{h}\right| \le \sum_{n\ge N+1} |a_n| \left|\frac{(z_0+h)^n - z_0^n}{h}\right| \le \sum_{n\ge N+1} |a_n|\, n \cdot r^{n-1}$$

$\exists N_1 > 0$. s.t $\forall n > N_1$ $\left|\frac{E_N(z_0+h) - E_N(z_0)}{h}\right| < \varepsilon$.

$\exists N_2 > 0$. s.t. $\forall n > N_2$ $|S_N'(z_0) - g(z_0)| < \varepsilon$.

Let $N = \max\{N_1, N_2\}$. $\exists \delta > 0$. s.t. $\forall |h| < \delta$. $\left|\frac{1}{h}(S_{N_h} - S_N) - S_N'\right| < \varepsilon$.

$\Rightarrow$ $f'(z_0) = g(z_0)$

Last, It's easy to check that $\limsup_{n\to\infty} |a_n|^{\frac{1}{n}} = \limsup_{n\to\infty} (n|a_n|)^{\frac{1}{n}}$.

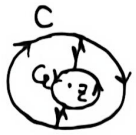
Def I.1.6. f defined on an open set Ω is said to be analytic at $z_0 \in \Omega$. Iff $\exists \sum_{n\ge 0} a_n (z - z_0)^n$ with positive convergence radius. s.t. $f(z) = \sum_{n\ge 0} a_n (z - z_0)^n$. $\forall z$ in a neighbourhood of z_0.

Coro I.1.7. By the I.1.6. we immediately have f is analytic on $\Omega \Rightarrow f$ is holomorphic on Ω

Thm I.1.8 (Cauchy's theorem) If a function f is holomorphic in an open set that contains a simple closed piecewise-smooth curve Γ and its interior, then $\int_\Gamma f = 0$

proof (Sketched) Using finite closed cubes A_k to approximate Int Γ $\square$

Thm I.1.9 (Cauchy Integral formula) If a function f is holomorphic on open set Ω containing a circle D. $\bar{D} \subseteq \Omega$ Let $C = \partial\Omega$ Then $f(z) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta \quad \forall z \in D$

Proof: 

$$\int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta \overset{\text{Cauchy}}{=\!=} \int_{C_\varepsilon} \frac{f(\zeta)}{\zeta - z}\,d\zeta = \int_{C_\varepsilon} \frac{f(\zeta)-f(z)}{\zeta - z}\,d\zeta + \int_{C_\varepsilon} \frac{f(z)}{\zeta - z}\,d\zeta$$

$$= 1\cdot A(\varepsilon) + \int_0^{2\pi} \frac{f(z)\; i\varepsilon e^{i\theta}}{\varepsilon e^{i\theta}}\,d\theta$$

$$= A(\varepsilon) + 2\pi i f(z) \quad \text{where } A(\varepsilon)\to 0 \text{ as } \varepsilon \to 0$$

thus $f(z) = \left(\int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta\right)\frac{1}{2\pi i}$ $\square$

Coro I.1.10. If f is holomorphic on open set Ω. then f have infinitely many derivatives. Moreover, if $C \subseteq \Omega$. (C is a circle), then $f^{(n)}(z) = \frac{n!}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}}\,d\zeta$. $\forall z \in C$

Proof: $n=1$, $\frac{f(z+h)-f(z)}{h} = \frac{1}{2\pi i h}\int_C \left(\frac{f(\zeta)}{\zeta - z - h} - \frac{f(\zeta)}{\zeta - z}\right)d\zeta = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta - z - h)(\zeta - z)}\,d\zeta$

let $h \to 0$ we have $f'(z) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta - z)^2}\,d\zeta$.

Assume n ✓ Now consider $n+1$ $\frac{f^{(n)}(z+h) - f^{(n)}(z)}{h} = \frac{n!}{2\pi i h}\int_C \frac{f(\zeta)}{(\zeta - z - h)^{n+1}} - \frac{f(\zeta)}{(\zeta - z)^{n+1}}\,d\zeta$

$$= \frac{n!}{2\pi i h}\int_C \frac{f(\zeta)\left[1 - \left(1 - \frac{h}{\zeta - z}\right)^{n+1}\right]}{(\zeta - z - h)^{n+1}}\,d\zeta$$

$$\xrightarrow{h\to 0} \frac{(n+1)!}{2\pi i}\int_C \frac{f(\zeta)\,d\zeta}{(\zeta - z)^{n+2}}$$ $\square$

Coro. I.1.11. f is holomorphic on open set Ω. then $\left|\frac{f^n(z)}{n!}\right| \leq \frac{\|f\|_C}{R^n}$ R is the radius of C

$z \in C \subseteq \Omega$, $\|f\|_C = \sup_{z\in C} |f(z)|$

proof: $\left|\frac{f^{(n)}(z)}{n!}\right| = \frac{1}{2\pi}\left|\int_0^{2\pi} \frac{f(z_0 + Re^{i\theta})}{(Re^{i\theta})^{n+1}}\, iRe^{i\theta}\right| d\theta \leq \frac{\|f\|_C}{R^n}$

Coro. I.1.12. f is holomorphic in an open set Ω, $z_0 \in D \subseteq \bar{D} \subseteq \Omega$ z_0 is the center of D

then $f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$ where $a_n = \frac{f^{(n)}(z)}{n!}$ $\forall z \in D$ Thus f is analytic on Ω

proof: $f(z) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta = \frac{1}{2\pi i}\int_C f(\zeta)\cdot\frac{1}{\zeta - z_0}\cdot\frac{1}{1 - \frac{z - z_0}{\zeta - z_0}}\,d\zeta$. where $C = \partial D$

$\forall z \in D \quad \left|\frac{z - z_0}{\zeta - z_0}\right| < 1 \Rightarrow \frac{1}{1 - \frac{z - z_0}{\zeta - z_0}} = \sum_{n=0}^{\infty}\left(\frac{z - z_0}{\zeta - z_0}\right)^n$ uniformly for $\zeta \in C$

$$\Rightarrow f(z) = \sum_{n=0}^{\infty} \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\,d\zeta\;(z - z_0)^n = \sum_{n=0}^{\infty} \frac{f^{(n)}(z)}{n!}(z - z_0)^n$$

Coro I. 1.13 (Liouville's thm): If f is entire and bounded. then f is constant

Thm I. 1.14. If f is a holomorphic function in a region Ω that vanishes on a sequence of distinct points with a limit point in Ω Then $f \equiv 0$

proof (sketched) $\{\omega_k\}_{k\geq 1}$, z_0 limit point Choose disc D containing z_0

Use the Taylor expansion at z_0 $\Rightarrow$ $\exists$ neighbourhood B s.t $f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n$

we have

Since z_0 is the limit point, $\forall |B \cap \{\omega_k\}_{k\geq 1}| = \infty$

If $f \not\equiv 0$ on B. then $\exists$ smallest $m \geq 0$ s.t. $a_m \neq 0$. $\Rightarrow f(z) = a_m (z-z_0)^m [1 + g(z-z_0)$

let $z = \omega_k$ then $1 + g(\omega_k - z_0) \to 1$. $a_m (\omega_k - z_0)^m \neq 0 \Rightarrow f(\omega_k) \neq 0$ contradiction

Thus $f \equiv 0$.

Next, let $U = \{z | f(z) = 0\}$, $z \in \Omega\}$ If $\{z_n\}_{n\geq 1} \to z_0 \subseteq U$. , by continuity $z_0 \in U \Rightarrow U$ closed

$\forall z \in U$. $\exists$ Neighbourhood B s.t. $B \subseteq U \Rightarrow U$ open

Since $z_0 \in U \subseteq \Omega$ connected we must have $\Omega = U$ □

Thm I. 1.15 (Morera's thm) f is continuous in the region Ω, $\forall$ closed curve D. we have $\int_D f(z)\,dz = 0$

Then f is holomorphic in Ω.

pf: $f(z)$ has a primitive $F(z)$. Namely, $F'(z) = f(z)$ By Coro. I. 12. $f(z)$ is holomorphic in Ω □

Coro. I. 1.16. (Avarage thm). f is holomorphic in the region Ω Then $\forall z \in \Omega$. $\exists \varepsilon > 0$. $\{\omega | |z-\omega| \leq \varepsilon\} \subseteq D$.

we have $f(z) = \frac{1}{2\pi} \int_0^{2\pi} f(z + \varepsilon e^{it})\,dt$

2. Meromorphic function, Laurent series and residue.

Def I. 2.1. 1) An isolated singularity of f is z_0 iff $\exists$ Neighbourhood B. s.t. ~~$\forall B \setminus \{z_0\}$ is holomorphic~~

f is defined on $B \setminus \{z_0\}$.

2) z_0 is a removable singularity iff $f \in H(\Omega \setminus \{z_0\})$, and we can define $f(z_0)$ s.t. $f \in H(\Omega)$

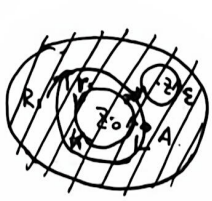
3) z_0 is a pole iff. $\frac{1}{f}$ has a zero at z_0, and is holomorphic in a neighbourhood of z_0

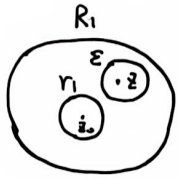
4) z_0 is an essential singularity iff z_0 is neither a removable singularity nor a pole

Thm I. 2.2. $z_0 \in \mathbb{C}$. f is holomorphic in $\{\omega | r < |z_0 - \omega| < R\}$. Then

(i) $C_k = \frac{1}{2\pi i} \int_{|z-z_0|=\rho} \frac{f(z)\,dz}{(z-z_0)^{k+1}}$ $k = 0, \pm 1, \pm 2, \ldots$

(ii). $\forall r < |z-z_0| < R$, we have $f(z) = \sum_{k=-\infty}^{\infty} C_k (z-z_0)^k$





proof: (i) $\frac{1}{2\pi i} \int_{|z-z_0|=\rho_1} \frac{f(z)\,dz}{(z-z_0)^{k+1}} = \frac{1}{2\pi i} \int_{|z-z_0|=\rho_2} \frac{f(z)\,dz}{(z-z_0)^{k+1}}$.

(ii). By Cauchy ~~thm~~ formula, $f(z) = \frac{1}{2\pi i} \int_{|\omega - z|=\varepsilon} \frac{f(\omega)\,d\omega}{\omega - z}$

By Cauchy thm, $\int_{|\omega - z_0|=R_1} \frac{f(\omega)\,d\omega}{\omega - z} = \int_{|\omega - z_0|=r_1} \frac{f(\omega)\,d\omega}{\omega - z} + \int_{|\omega - z|=\varepsilon} \frac{f(\omega)\,d\omega}{\omega - z}$

) $\Rightarrow 2\pi i f(z) = \int_{|\omega - z_0| = R_1} \frac{f(\omega)\,d\omega}{\omega - z} - \int_{|\omega - z_0| = r_1} \frac{f(\omega)\,d\omega}{\omega - z}$

When $|\omega - z_0| = R_1 > |z - z_0|$

$$\frac{1}{\omega - z} = \frac{1}{(\omega - z_0) - (z - z_0)} = \frac{1}{1 - \frac{z - z_0}{\omega - z_0}} = \frac{1}{\omega - z_0} \sum_{k=0}^{\infty} \left(\frac{z - z_0}{\omega - z_0}\right)^k \quad \text{uniformly for } |\omega - z_0| = R_1$$

When $|\omega - z_0| = r_1 < |z - z_0| \Rightarrow \frac{-1}{\omega - z} = \frac{1}{z - z_0} \frac{1}{1 - \frac{\omega - z_0}{z - z_0}} = \frac{1}{z - z_0} \sum_{k=0}^{\infty} \left(\frac{\omega - z_0}{z - z_0}\right)^k$ uniformly for $|\omega - z_0| = r_1$

Then $f(z) = \sum_{k=0}^{\infty} \frac{1}{2\pi i} \int_{|\omega - z_0| = R_1} \frac{f(\omega)\,d\omega}{(\omega - z_0)^{k+1}} (z - z_0)^k + \sum_{k=-\infty}^{-1} \frac{1}{2\pi i} \int_{|\omega - z_0| = r_1} \frac{f(\omega)\,d\omega}{(\omega - z_0)^{k+1}} (z - z_0)^k$

$= \sum_{k=-\infty}^{\infty} c_k (z - z_0)^k$ □

Rmk. 1) When $\forall k < 0.\ c_k = 0$ then z_0 is a removable singularity.

2) When $\forall k \le -n\ \ c_k = 0$, $c_{-n} \neq 0$. then z_0 is an n-th pole.

3) When $\{k < 0 \mid c_k \neq 0\}$ is infinite, then z_0 is an essential singularity.

4) z_0 is a removable iff $\exists$ Neighbourhood B s.t. $|f(z)| \le M.\ \forall z \in B$

5) z_0 is an n-th pole iff $\exists$ Neighbourhood B. s.t. $g(z) = f(z)\cdot(z - z_0)^n$ is holomorphic in B

$\&\ g(z) \neq 0$ ①

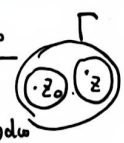
iff $\lim_{z \to z_0} |f(z)| = \infty$ ②

6). f is holomorphic in $B \setminus \{z_0\}$., z_0 is an essential singularity, then $\mathrm{Im} f$ is dense in $\mathbb{C}_\infty$.

proof of 5), ② only to prove ⇐ : let $g(z) = \begin{cases} \frac{1}{f(z)} & z \in B \setminus \{z_0\} \\ 0 & z = z_0 \end{cases}$

then $g(z)$ is holomorphic in $B \setminus \{z_0\}$, continuous at z_0

Then $\forall z \in B \setminus \{z_0\}$. $\int_\Gamma \frac{g(\omega)\,d\omega}{\omega - z} = \int_{|\omega - z_0| = r} \frac{g(\omega)\,d\omega}{\omega - z} + \int_{|\omega - z_0| = r} \frac{g(\omega)\,d\omega}{\omega - z}$



$= \int_{|\omega - z_0| = r} \frac{g(\omega)\,d\omega}{(\omega - z)} + 2\pi i g(z)$ (Cauchy formula)

$= \int_0^{2\pi} \frac{g(z_0 + re^{it})\, r \cdot i t\, e^{it}\, dt}{z_0 + re^{it} - z} \to 0.\ (r \to 0)$

Thus $\forall z \in B \setminus \{z_0\}.\ \int_\Gamma \frac{g(\omega)\,d\omega}{\omega - z} = 2\pi i g(z)$

We can prove that if $g \in C(B)$ then $\varphi(z) = \int_\Gamma \frac{g(\omega)}{\omega - z}\,d\omega$ is holomorphic in D.

By continuity, $g(z)$ is holomorphic at z_0 (analytic)

Thus. $\exists n$. s.t. $g(z) = (z - z_0)^n h(z)$ $h(z)$ is analytic in B $h(z_0) \neq 0$

$\Rightarrow f(z) = \frac{1}{(z - z_0)^n} \frac{1}{h(z)}$ By ② we have $f(z)$ has an n-th pole z_0

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6) ① Suppose $\exists\, \omega \in \mathbb{C}_\infty$ $\exists\, \varepsilon > 0, \exists\, \delta > 0$ $\forall\, |z - z_0| < \delta$ we have $|f(z) - \omega| \geq \varepsilon$

$\Rightarrow \frac{1}{f(z)-\omega}$ is holomorphic & bounded in $V(z_0, \delta) \setminus \{z_0\}$.

$\Rightarrow z_0$ is a removable for $\frac{1}{f(z)-\omega}$. $\Rightarrow \frac{1}{f(z)-\omega} = \sum_{k=0}^{\infty} a_k (z - z_0)^k$

$\Rightarrow f(z) = \frac{1}{g(z)} + \omega = \frac{1}{(z-z_0)^n h(z)} + \omega \Rightarrow z_0$ is not an essential singularity.

② $\omega = \infty$ If not, then $f(z)$ is bounded. $\Rightarrow z_0$ is removable. □

Def I.2.3. Let z_0 be an isolated singularity of f $f = \sum_{k=-\infty}^{\infty} c_k (z - z_0)^k$

$c_{-1} = \frac{1}{2\pi i} \int_\Gamma f(z)\,dz$ is called the residue of f at z_0 denoted by $\mathrm{Res}(f, z_0)$.

ThmI.2.4. f is holomorphic in $C \setminus \{z_0\}$ C is a region containing z_0.

Then $\int_C f(z)\,dz = 2\pi i\, \mathrm{Res}(f, z_0)$

Pf: $\int_C f(z)\,dz = \int_{C_\varepsilon} f(z)\,dz$. $= \int_{C_\varepsilon} \frac{a_{-1}}{z - z_0}\,dz$ (Cauchy formula)

$= 2\pi i a_{-1} = 2\pi i\, \mathrm{Res}(f, z_0)$ □.

ThmI.2.5. f has a pole of order n at z_0, then $\mathrm{Res}(f, z_0) = \lim_{z \to z_0} \frac{1}{(n-1)!} \left(\frac{d}{dz}\right)^{n-1} (z - z_0)^n f(z)$

Pf: Since $(z - z_0)^n f(z) = a_{-n} + a_{-n+1}(z - z_0) + \cdots + a_{-1}(z - z_0)^{n-1} + G(z)(z - z_0)^n$

$\Rightarrow a_{-1} = \lim_{z \to z_0} \frac{1}{(n-1)!} \frac{d^{n-1}}{dz} (z - z_0)^n f(z)$ □.

Def I.2.6. f is a meromophic function in region Ω. iff $\exists\, z_0, \ldots, z_n, \ldots$ be distinct & isolated, s.t. f is holomorphic in $\Omega \setminus \{z_n\}_{n \geq 0}$, $\{z_n\}_{n \geq 0}$ be poles.

ThmI.2.6. Suppose f is a meromophic function in Ω. poles be $z_1, \ldots, z_N$, then

$\int_{\partial\Omega} f(z)\,dz = 2\pi i \sum_{k=1}^{N} \mathrm{Res}(f, z_k)$

Thm I.2.7. (Argument Principle). f is meromorphic in ~~an open set~~ region Ω. If f has no poles and zeros on $\partial\Omega$, Then $\frac{1}{2\pi i} \int_{\partial\Omega} \frac{f'(z)}{f(z)}\,dz$ = numbers of zeros in Ω − numbers of poles

proof: If f has a zero z_0 of order n.

$\Rightarrow f(z) = (z - z_0)^n g(z) \Rightarrow \frac{f'(z)}{f(z)} = \frac{n}{z - z_0} + \frac{g'(z)}{g(z)}$ is meromorphic near z_0. ($\frac{g'(z)}{g(z)}$ holomorphic near z_0)

$\Rightarrow \int_{C_\varepsilon} \frac{f'(z)}{f(z)}\,dz = 2\pi i\, n$

Similarly, if f has a pole z_0 of order n

$\Rightarrow f(z) = (z - z_0)^{-n} h(z) \Rightarrow \int_{C'_\varepsilon} \frac{f'(z)}{f(z)}\,dz = -2\pi i\, n$

Adding them by Cauchy theorem. we have the desired result. □

Thm I.2.8 (Rouché thm) f, g be holomorphic in region Ω If $|f(z)| > |g(z)|$ $\forall z \in \partial\Omega$

Then f & $g+f$ have the same number of zeros in Ω

proof: $\forall t \in [0,1]$ define $f_t(z) = f(z) + tg(z)$. let n_t be the number of zeros of f_t in Ω

By $|f(z)| > |g(z)|$ we have $|f_t(z)| \geq |f(z)| - |g(z)| > 0$ $\forall z \in \partial\Omega$

By Thm I.2.7. $n_t = \frac{1}{2\pi i}\int_{\partial\Omega} \frac{f_t'(z)}{f_t(z)}\,dz$ $\Rightarrow$ n_t is continuous in $[0,1]$

Since $n_t \in \mathbb{Z}$ $\Rightarrow$ $n_0 = n_1$ $\square$.

Thm I.2.9 (Open mapping thm). If f is holomorphic & non-constant in region Ω. then the map f is open

proof: $\forall w_0 \in \mathrm{Im} f$, $\exists z_0 \in \Omega$. $f(z_0) = w_0$ $\forall w$ near w_0, we prove that $w \in \mathrm{Im} f$

Let $g(z) = f(z) - w$ $\quad g(z) = f(z) - w_0 + w_0 - w = F(z) + G(z)$



Since f is non-constant

$\exists \delta > 0$, s.t. $\forall |z - z_0| = \delta$. $f(z) \neq w_0$.

$\exists \varepsilon > 0$. s.t $|f(z) - w_0| \geq \varepsilon$. on $|z - z_0| = \delta$.

If $|w_0 - w| < \varepsilon$, then by Rouché's thm, $(|f(z)| > |G(z)|)$ $\Rightarrow$ $g(z)$ has at least on zero $\square$

Thm I.2.10 (Maximum modulus principle) If f is a non-constant & holomorphic function in region Ω, then $|f(z)|$ cannot attain a maximum in Ω



proof: Suppose not, then $|f(z_0)| \geq |f(z)|$ $\forall z \in \Omega$, $z_0 \in \Omega$

By Thm I.2.9, we immediately have $\exists$ A containing $f(z_0)$ s.t. $A \subseteq \mathrm{Im} f$

$\Rightarrow \exists z' \in A$. s.t $|f(z')| > |f(z_0)|$ contradiction ! $\square$.

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§3. Riemann mapping theorem ($\mathbb{D}$ be the disc: $\{z\in\mathbb{C}\mid |z|<1\}$)

Some special cases for $\mathrm{Aut}(\mathbb{D})$

Lemma I.3.1 (Schwarz) $f:\mathbb{D}\to\mathbb{D}$ holomorphic $f(0)=0$. Then

(i) $\forall z\in\mathbb{D}\quad |f(z)|\le|z|$

(ii) If $\exists z_0\neq 0$ s.t. $|f(z_0)|=|z_0|$, then f is a rotation

(iii) $|f'(0)|\le 1$ and if $|f'(0)|=1$, then f is a rotation.

proof: $f(z)=\sum_{k\ge 0}^{\infty} a_k z^k$. ($a_0=0$) convergents uniformly wherever in $|z|\le 1$.

Then $\frac{f(z)}{z}$ is holomorphic in $\mathbb{D}$. If $|z|=r<1$. then $\left|\frac{f(z)}{z}\right|\le\frac{1}{r}$.

By maximum modulus principle, $\forall |z|<r$, $\left|\frac{f(z)}{z}\right|<\frac{1}{r}$.

let $r\to 1^-$ we have $\left|\frac{f(z)}{z}\right|\le 1$ i).

ii) $\exists z_0\neq 0\in\mathbb{D}$. s.t. $\left|\frac{f(z_0)}{z_0}\right|=1$. then by maximum modulus principle.

$\frac{f(z)}{z}$ is constant c where $|c|=1$ let $c=e^{i\theta}$ then $f(z)=e^{i\theta}z$

iii). $|f'(0)|=\left|\lim_{z\to 0}\frac{f(z)}{z}\right|\le 1$ when $|f'(0)|=1$ by m.m.p... □

Rmk. If just using Cauchy's theorem, we define $g(z)=\begin{cases}\frac{f(z)}{z} & 0<|z|<1.\\ f'(0) & |z|=0\end{cases}$

$g(z)\in H(\mathbb{D}\setminus\{0\})$. $g(z)\in C(\mathbb{D})$ By the previous theorem $g(z)\in H(\mathbb{D})$.

Then use m.m.p. □

Def I.3.2. A map $f:U\to V$ is called the conformal map iff f is holomorphic & bijective.

A conformal map from open set Ω to itself forms the group $\mathrm{Aut}(\Omega)$.

Thm I.3.3. If $f\in\mathrm{Aut}(\mathbb{D})$, then $\exists\theta\in\mathbb{R}$, $\alpha\in\mathbb{D}$ s.t. $f(z)=e^{i\theta}\frac{\alpha-z}{1-\bar{\alpha}z}$

proof: let $\alpha\in\mathbb{D}$ s.t. $f(\alpha)=0$. , $\psi_\alpha:\mathbb{D}\to\mathbb{D}$. $z\mapsto\frac{\alpha-z}{1-\bar{\alpha}z}$

then $|\psi_\alpha(e^{i\theta})|=\left|\frac{-e^{i\theta}+\alpha}{e^{i\theta}(e^{-i\theta}-\bar{\alpha})}\right|=1$ By m.m.p. $|\psi_\alpha(z)|<1\quad\forall z\in\mathbb{D}$.

Also $\psi_\alpha\circ\psi_\alpha(z)=z$. So $\psi_\alpha\in\mathrm{Aut}(\mathbb{D})$.

let $g(z)=f\circ\psi_\alpha(z)$, then $g(0)=f\circ\psi_\alpha(0)=0$

By Schwarz lemma, $\begin{cases}|g(z)|\le|z| & \forall z\in\mathbb{D}\\ |g^{-1}(w)|\le|w| & \forall w\in\mathbb{D}.\quad (g^{-1}(0)=0)\end{cases}$

let $w=g(z)\Rightarrow |g(z)|=|z|$ So $g(z)=e^{i\theta}z$

Thus $f(z)=e^{i\theta}\psi_\alpha(z)$ □

Coro. I.3.4. $f\in\mathrm{Aut}(\mathbb{D})$, $f(0)=0$ Then f is a rotation

2nd. $\mathrm{Aut}(\mathbb{H})$.

Thm I.3.5. $f \in \mathrm{Aut}(\mathbb{H})$ if and only if f takes the form $f_M: \frac{az+b}{cz+d}$, $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{R})$

proof: 1) If $M \in SL_2(\mathbb{R})$, then $\mathrm{Im}(f_M(z)) = \frac{\mathrm{Im}(z)}{|cz+d|^2} > 0 \quad \forall z \in \mathbb{H} \Rightarrow \mathrm{Im} f_M \subseteq \mathbb{H}$

2) $f_M \circ f_{M'} = f_{MM'}$, thus $(f_M)^{-1} = f_{M^{-1}}$

3). $\mathrm{Aut}(\mathbb{H})$ acts transitively. $\forall z, w \in \mathbb{H}$

W.L.O.G. take $w = i$ (Using 2)) Choose $c \in \mathbb{R}$ s.t. $|cz|^2 = \mathrm{Im}(z)$

Choose $M_1 = \begin{pmatrix} 0 & -c^{-1} \\ c & 0 \end{pmatrix}$ then $\mathrm{Im}(f_{M_1}(z)) = \frac{\mathrm{Im}(z)}{|cz|^2}$

Next choose $M_2 = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$ $b \in \mathbb{R}$ s.t. $b = -\mathrm{Re}(f_{M_1}(z))$

then $\mathrm{Im}(f_{M_2M_1}(z)) = i \quad \mathrm{Re}(f_{M_2M_1}(z)) = 0$ take $M = M_2M_1$

4) $F: \mathbb{H} \to \mathbb{D} \quad w \to \frac{i-w}{i+w}$ is a conformal map. when $F^{-1}(z) = i\frac{1-z}{1+z}$

$M_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ $(\theta \in \mathbb{R})$ Then $F \circ f_{M_\theta} \circ F^{-1}(z) = e^{-2i\theta} F(z)$.

5). Suppose $f \in \mathrm{Aut}(\mathbb{H})$ $\exists!\, \beta \in \mathbb{H}$ s.t. $f(\beta) = i$.

$\exists N \in SL_2(\mathbb{R})$ s.t. $f_N(i) = \beta$. Then $g = f \circ f_N$, $g(i) = i$.

$\Rightarrow F \circ g \circ F^{-1} \in \mathrm{Aut}(\mathbb{D})$ where $F \circ g \circ F^{-1}(0) = 0$.

By Coro I.3.4 $F \circ g \circ F^{-1}(z) = e^{i\theta} z \Rightarrow g = F^{-1} \circ \varphi_{\theta_1} \circ F = f_{M_{\theta_2}}$ (By 4))

Thus $f = f_{M_{\theta_2} N^{-1}} \in SL_2(\mathbb{R})$ $\square$

Rmk. Notice that $f_M = f_{-M}$ thus $\mathrm{Aut}(\mathbb{H}) \cong PSL_2(\mathbb{R})$

< Thm I.3.6 (Riemann mapping theorem) Ω is proper (not-empty and not the whole of $\mathbb{C}$) and simply connected. If $z_0 \in \Omega$, then $\exists!$ conformal map $F: \Omega \to \mathbb{D}$ s.t. $F(z_0) = 0$ & $F'(z_0) > 0$.

Before the proof, we need the technical tool: Montel's theorem

Def I.3.7. 1) $\mathcal{F}$ is a family of function in $H(\Omega)$, Ω open in $\mathbb{C}$. $\mathcal{F}$ is <u>normal</u> iff every sequence in $\mathcal{F}$ has a subsequence that converges uniformly on every compact set of Ω.

2) $\mathcal{F}$ is <u>uniformly bounded</u> on compact subsets of Ω iff $\forall$ compact $K \subseteq \Omega$. $\exists B > 0$ $|f(z)| \leq B$, $\forall z \in B$, $f \in \mathcal{F}$.

3) $\mathcal{F}$ is <u>equicontinuous</u> on compact K iff $\forall \varepsilon > 0$ $\exists \delta > 0$. $\forall |z-w| < \delta$. then $|f(z)-f(w)| < \varepsilon$ $\forall f \in \mathcal{F}$.

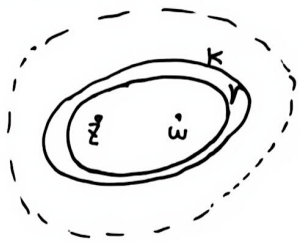
Thm I.3.8 (Montel) $\mathcal{F} \subseteq H(\Omega)$ is uniformly bounded on compact subsets of Ω

Then i) $\mathcal{F}$ is equicontinuous on every compact subset of Ω.

ii) $\mathcal{F}$ is a normal family.

Proof: i). $K \subseteq \Omega$ compact choose $r > 0$ s.t. $\forall z \in K$. s.t. $\overline{V}(z, 3r) \subseteq \Omega$

Let $|z-w| < r$, $z, w \in K$, $\gamma = \partial D_{2r}(w)$



Then $f(z) - f(w) = \frac{1}{2\pi i} \int_\gamma f(\zeta) \left(\frac{1}{\zeta - z} - \frac{1}{\zeta - w} \right) d\zeta$.

where $\left| \frac{1}{\zeta - z} - \frac{1}{\zeta - w} \right| = \frac{|z-w|}{|\zeta - z||\zeta - w|} \leq \frac{|z-w|}{r^2}$

Thus $|f(z) - f(w)| \leq \frac{1}{r} \sup_{\substack{z \in K \\ f \in \mathcal{F}}} |f(z)| \cdot |z-w|$. $\forall f \in \mathcal{F}$. $\Rightarrow \mathcal{F}$ is equicontinuous on compa sets of Ω $\square$.

For ii) & i) & uniformly bounded on compact subsets of Ω, use Arzela-Ascoli theorem $\square$

We can now start our proof:

① First we show that $\Omega \approx$ an open subset of the unit disc that contcains the origin.

Choose $\alpha \in \mathbb{C} \setminus \Omega$ such that $f(z) = \log(z - \alpha)$ is defined on the branch of sheet

$\Rightarrow e^{f(z)} = z - \alpha$ Thus f is injective.

Besides, choose $w \in \Omega$, Assume $\exists \{z_n\} \subseteq \Omega$, s.t. $f(z_n) \to f(w) + 2\pi i \Rightarrow z_n \to w \cdot \Rightarrow f(z_n) \to f(w)$

contradiction! Thus $F(z) = \frac{1}{f(z) - (f(w) + 2\pi i)}$ $\Omega \to F(\Omega)$ is holomorphic, & bijective.

Also $F(\Omega)$ is bounded. Then translate and rescale we can make $F(\Omega) \subseteq \mathbb{D}$. and contains th origin.

② Let Ω be an open subset of $\mathbb{D}$ with $0 \in \Omega$.

$\mathcal{F} = \{ f : \Omega \to \mathbb{D}$ holomorphic, injective and $f(0) = 0 \}$.

Let $s = \sup |f'(0)|$.

Choose $\{f_n\} \subseteq \mathcal{F}$ s.t. $|f_n(0)| \to s$ By Montel's thm $\exists \{f_{n_k}\}$ converges uniformly on compact sets to a holomorphic f on Ω f is not constant

Suppose f is not injective $\exists z_1 \neq z_2$, $f(z_1) = f(z_2)$

Def. $g_n(z) = f_n(z) - f(z_1)$ $\rightsquigarrow$ $g(z) = f(z) - f(z_1)$ Since g is not constant

Then $g(z)$ has an isolated zero at z_2 is a small disc

By Argument principle, $\frac{1}{2\pi i}\int_{\partial\gamma} \frac{g'(z)dz}{g(z)} = 1$ γ centered at z_2

where $\frac{1}{g_n(z)} \to \frac{1}{g(z)}$ on γ, $g_n'(z) \to g'(z)$ on γ $\Rightarrow \frac{1}{2\pi i}\int_\gamma \frac{g_n'(z)}{g_n(z)}\,dz \ (=0) \to 1$

Contradiction. Thus f is injective.

By continuity & m.m.p. We have $|f(z)| < 1$, $f(0) = 0$ $\Rightarrow f \in \mathcal{F}$, $|f'(0)| = s$

③ We show that $f: \Omega \to \mathbb{D}$ is a conformal map.

It suffices to show that f is surjective

Assume that $\exists \alpha \in \mathbb{D}$. s.t. $\forall z \in \Omega$. $f(z) \neq \alpha$

Consider $\psi_\alpha \in \mathrm{Aut}(\mathbb{D})$. s.t. $\psi_\alpha(\alpha) = 0$. : $\psi_\alpha(z) = \frac{\alpha - z}{1 - \bar{\alpha} z}$.

Since Ω is simply connected, so is $U = (\psi_\alpha \circ f)(\Omega)$. which doesn't contain the origin.

Then we can define $g(w) = e^{\frac{1}{2}\log w}$. on U.

Consider $F = \psi_{g(\alpha)} \circ g \circ \psi_\alpha \circ f$. $F(0) = 0$. F is injective and maps into $\mathbb{D}$

Let $h(w) = w^2$. $\Rightarrow f = \psi_\alpha^{-1} \circ h \circ \psi_{g(\alpha)}^{-1} \circ F = \Phi \circ F$. where Φ is not injective. since h is not

$\Rightarrow$ By Schwarz lemma, $|\Phi'(0)| < 1$ $\Rightarrow$ $|f'(0)| < |F'(0)|$ Contradiction! $\square$